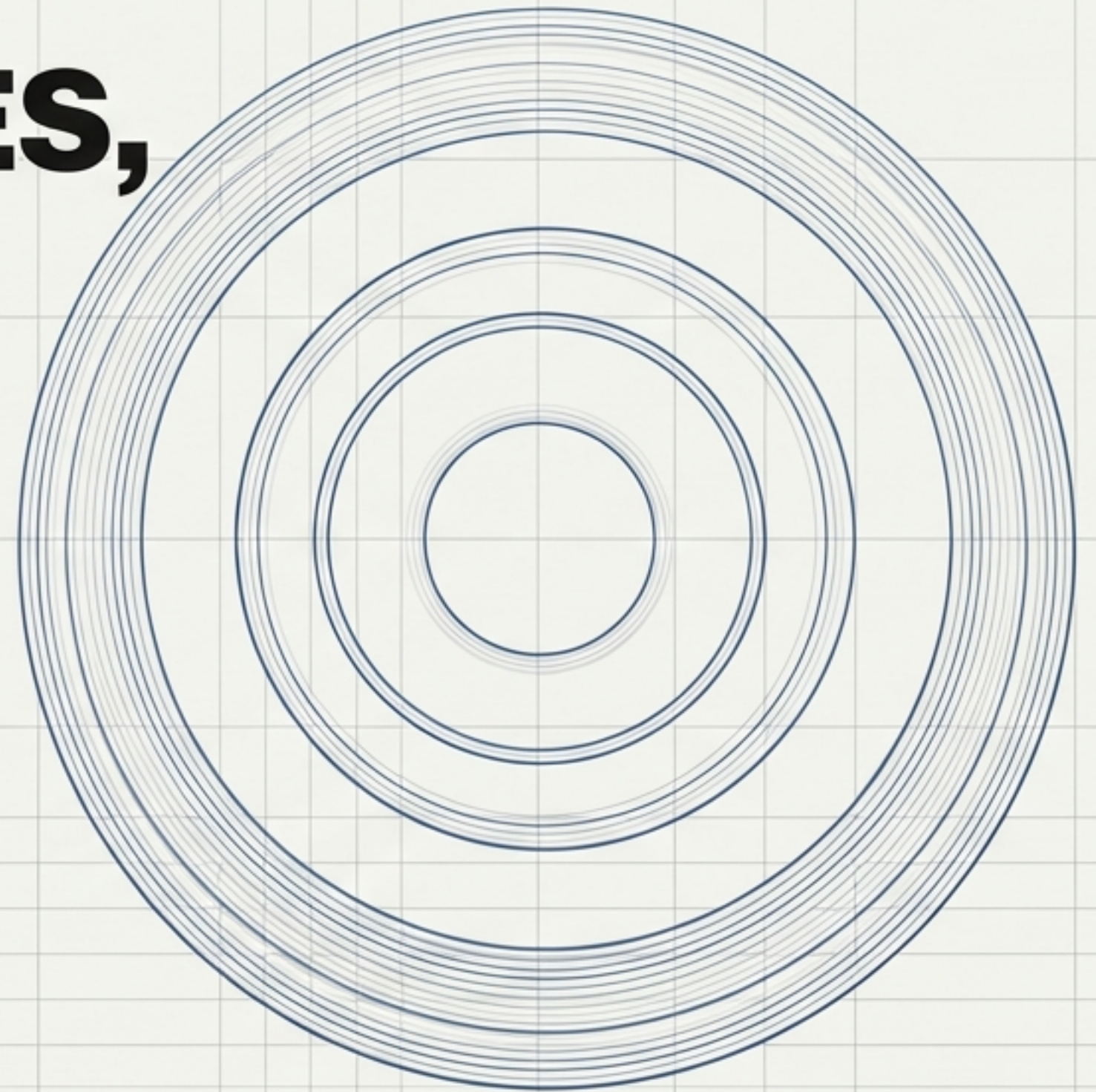


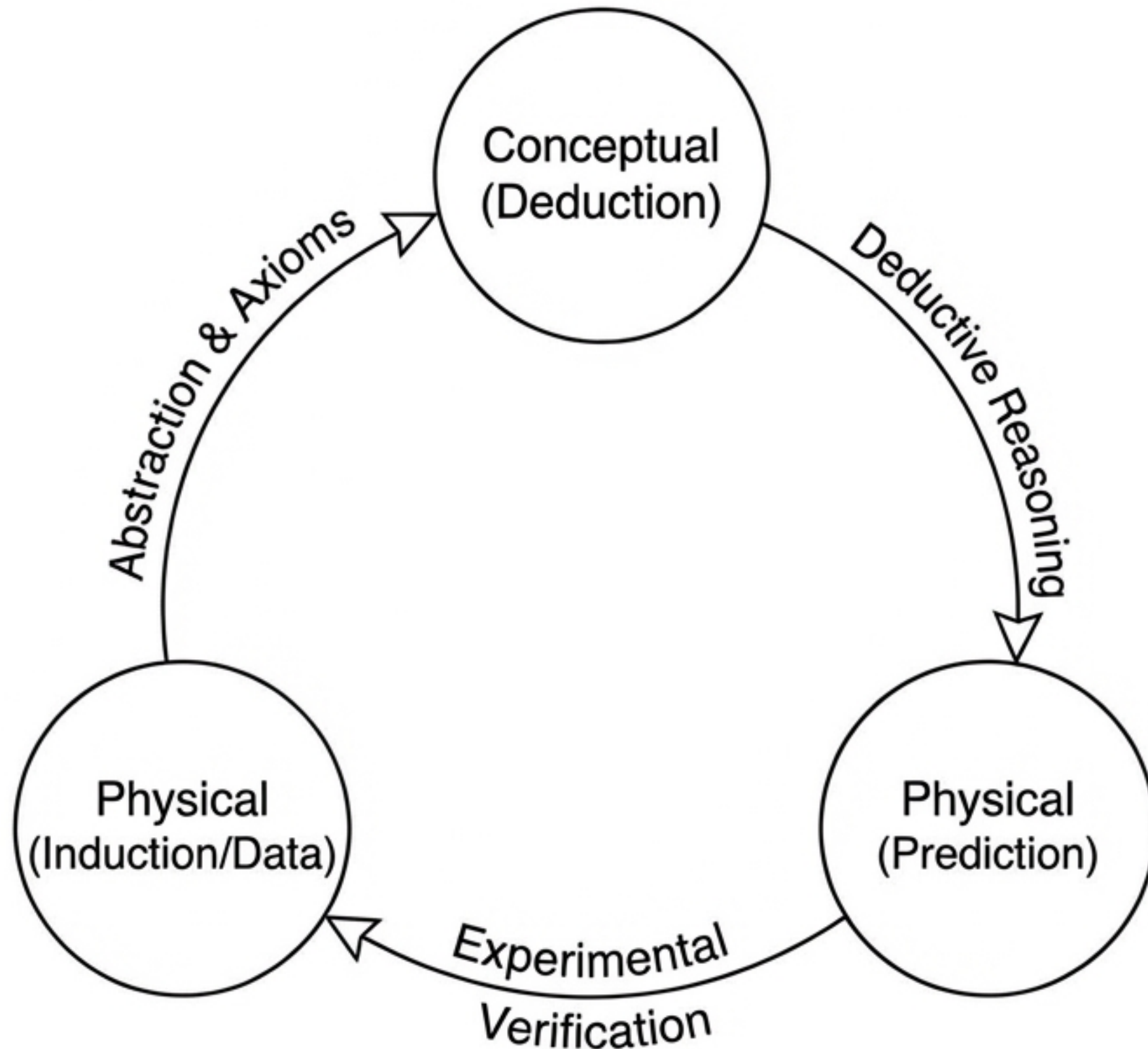
PROBABILITY, RANDOM VARIABLES, AND STOCHASTIC PROCESSES

A Deductive Framework for
Modeling Uncertainty

Based on the classic text by
Athanasios Papoulis (Third Edition)



The Modeling Cycle: Observation, Deduction, Prediction



Probability is a mathematical discipline; applying it to the real world requires inductive reasoning, but the theory itself is strictly deductive.”

Three Competing Definitions of Probability

Axiomatic (The Standard)

- Based on Set Theory and three postulates.
- Strength: Deductive, rigorous, avoids conceptual ambiguity.
- Usage: The method used in this presentation.

Relative Frequency

$$P(A) = \lim_{n \rightarrow \infty} \frac{n_A}{n}$$

- Based on empirical observation of mass phenomena.
- Critique: n is never truly infinite in the physical world; useful for application, not definition.

Classical Definition

$$P(A) = \frac{N_A}{N}$$

- Ratio of favorable outcomes (N_A) to total outcomes (N).
- Critique: Relies on the circular assumption of “equally likely” outcomes.

The Flaw in the Classical Definition

Why “Common Sense” is mathematically circular.

The Argument:

- **The Classical Definition:** $P(A) = \frac{N_A}{N}$
- **Condition:** Provided all outcomes are “equally likely”.
- **The Logical Flaw:** “Equally likely” is synonymous with “equally probable”. We are using probability to define probability.



Total Outcomes (N) = 6

Scenario: A Loaded Die (Biased)

If the die is weighted so ‘6’ appears 20% of the time:

- Favorable Outcomes for rolling a six (N_A) = 1
- Total Outcomes (N) = 6
- **Classical Prediction:** $P(6) = 1/6$ (16.6%)
- **Actual Reality:** $P(6) = 0.20$ (20%)

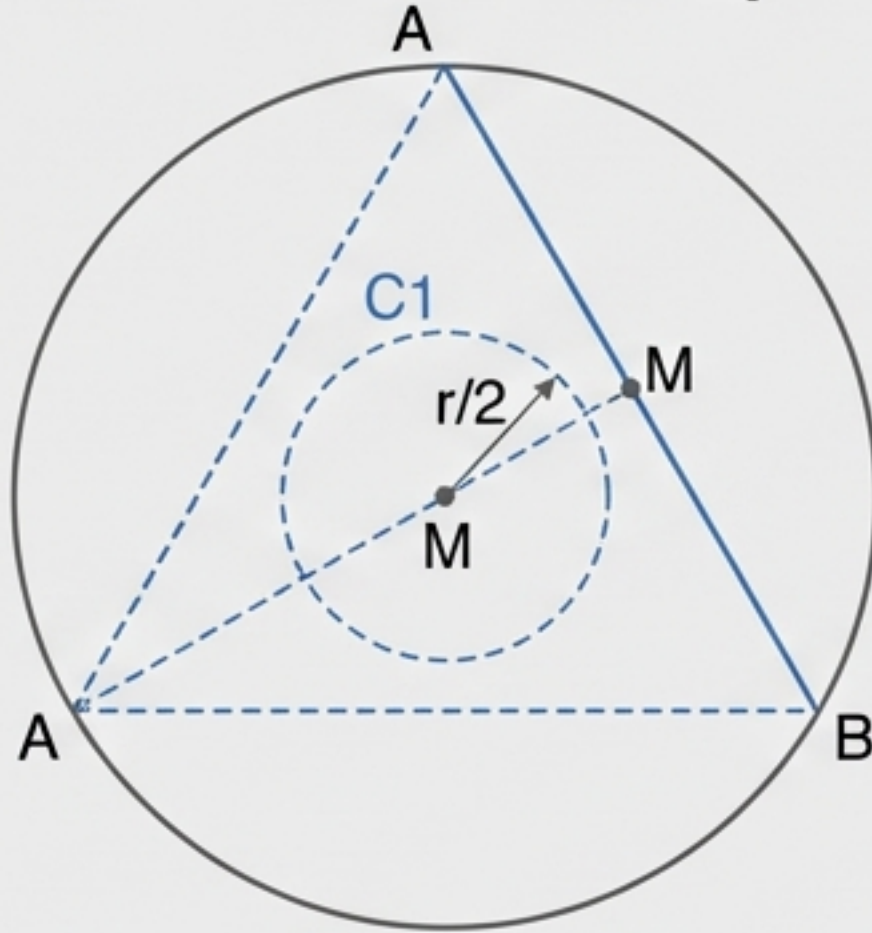
Conclusion: The classical definition cannot model asymmetric or biased systems.
(Reference the loaded die critique.)

The Ambiguity of “Random” (Bertrand’s Paradox)

Without a precise probability space, “random” is meaningless.

Problem: Draw a chord at random in a circle. What is the probability its length exceeds the side of an inscribed equilateral triangle?

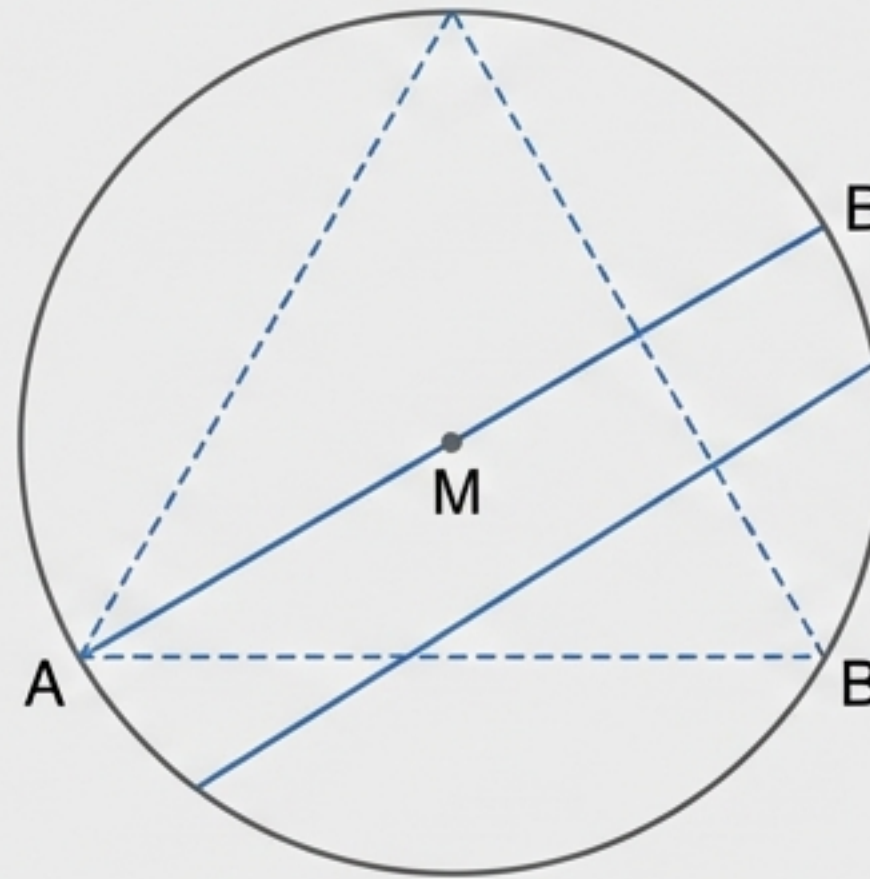
Solution 1: Random Endpoints



Rotate the chord around a fixed point on the circumference.

$$P = 1/3$$

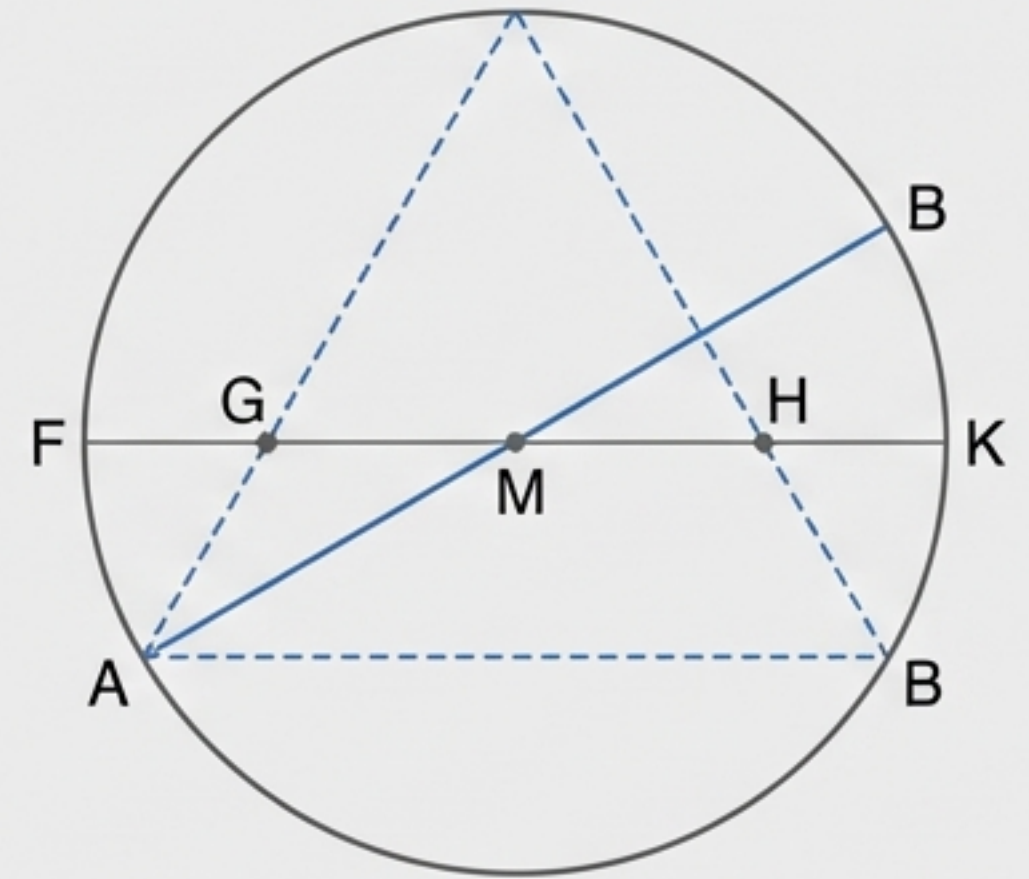
Solution 2: Random Radius



Move the chord parallel to a tangent.

$$P = 1/2$$

Solution 3: Random Center



Pick a random point within the circle area as the chord's center.

$$P = 1/4$$

Set Theory: The Language of Events

Definitions:

- The Space (S): The set of all experimental outcomes (The Certain Event).
- The Elements (ζ): Single outcomes or objects within the set.
- The Impossible Event ($\emptyset$): The set containing no elements.

Notation & Example:

Notation:

$\zeta \in A$ (ζ is an element of set A)

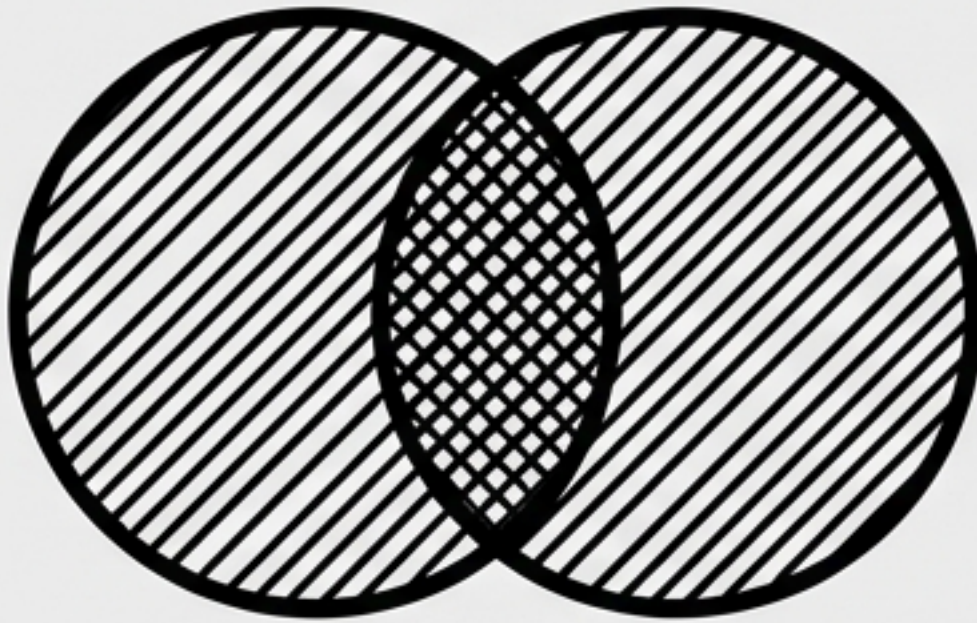
$A \subset B$ (A is a subset of B)

Example: Rolling a Die

$$\left\{ \begin{array}{l} S = \{1, 2, 3, 4, 5, 6\} \\ \text{Event "Even"} = \{2, 4, 6\} \end{array} \right\}$$

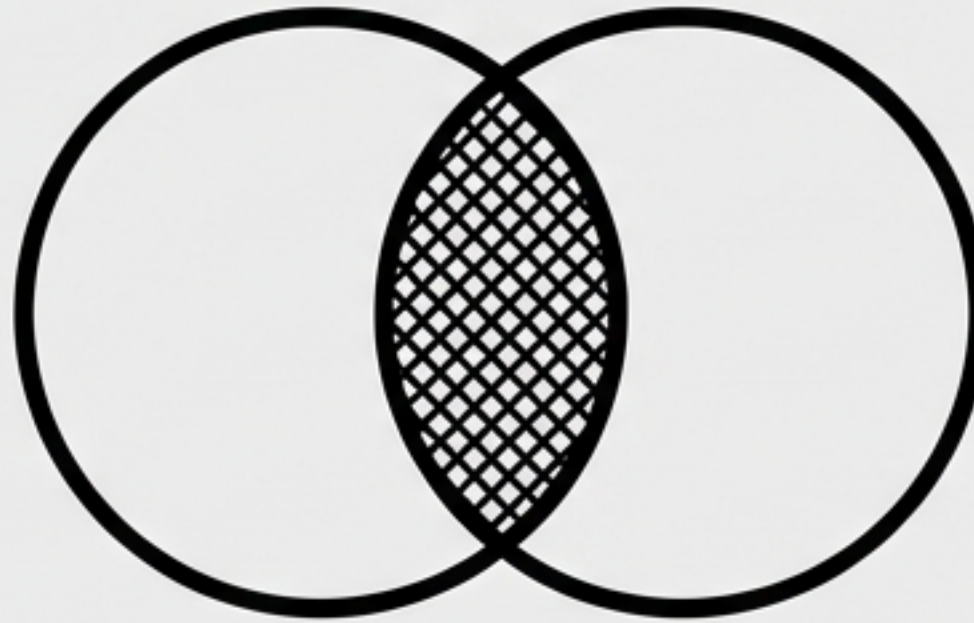
Visualizing Uncertainty

Union ($A + B$)



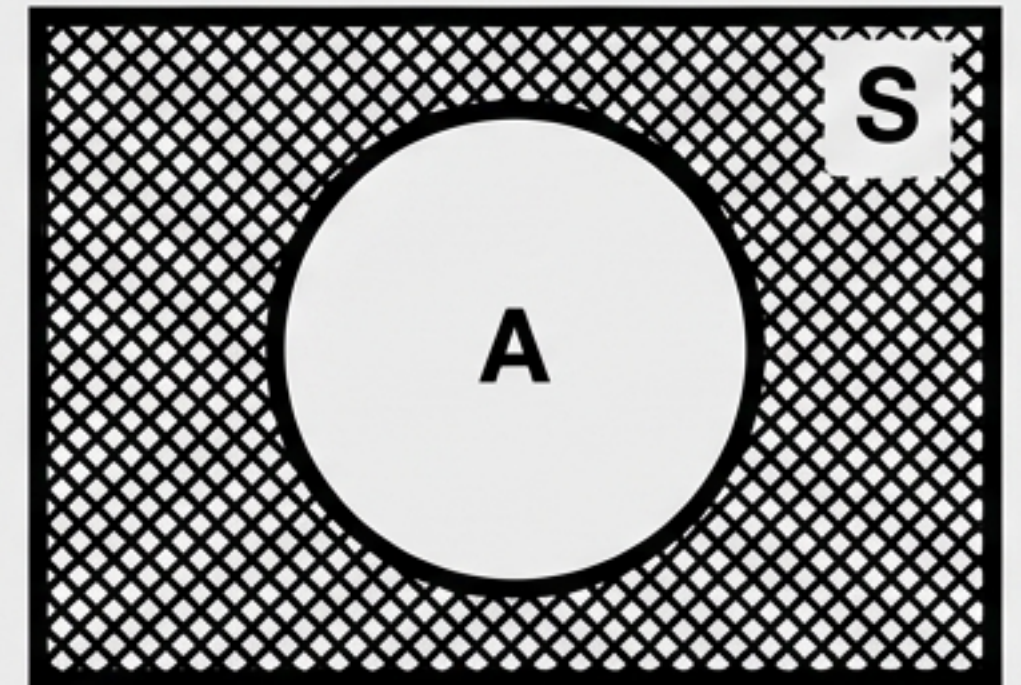
Occurs if A OR B
(or both) occur.

Intersection (AB)



Occurs if BOTH
A AND B occur.

Complement ($\bar{A}$)



Occurs if A does
NOT occur.

The Probability Space

The Trinity of Mathematical Modeling

The Space (S)

The set of all possible experimental outcomes.

The Field of Events (F)

A Borel field of subsets of S. Not just any list, but a structured collection closed under complements and unions.

The Probability Measure (P)

A number assigned to each event in F.



Why a Field? To strictly handle infinite possibilities (like points on a line) without creating mathematical paradoxes.

The Three Axioms of Probability

Formalized by A. Kolmogorov (1933)

I. Non-Negativity

$$P(A) \geq 0$$

II. Normalization

$$P(S) = 1$$

III. Additivity

If events A and B are mutually exclusive ($AB = \emptyset$):

$$P(A + B) = P(A) + P(B)$$

Immediate Consequences

Deducing truth from the axioms.

1. The Impossible Event

$$P(\emptyset) = 0$$

Since $A + \emptyset = A$, and A and $\emptyset$ are mutually exclusive, by Axiom III, $P(A) = P(A + \emptyset) = P(A) + P(\emptyset)$. Subtracting $P(A)$ from both sides yields $P(\emptyset) = 0$.

2. Complements

$$P(\bar{A}) = 1 - P(A)$$

Since $A + \bar{A} = S$ and they are mutually exclusive, $P(S) = P(A) + P(\bar{A})$. By Axiom II, $P(S) = 1$. Thus, $1 = P(A) + P(\bar{A})$, which leads to $P(\bar{A}) = 1 - P(A)$.

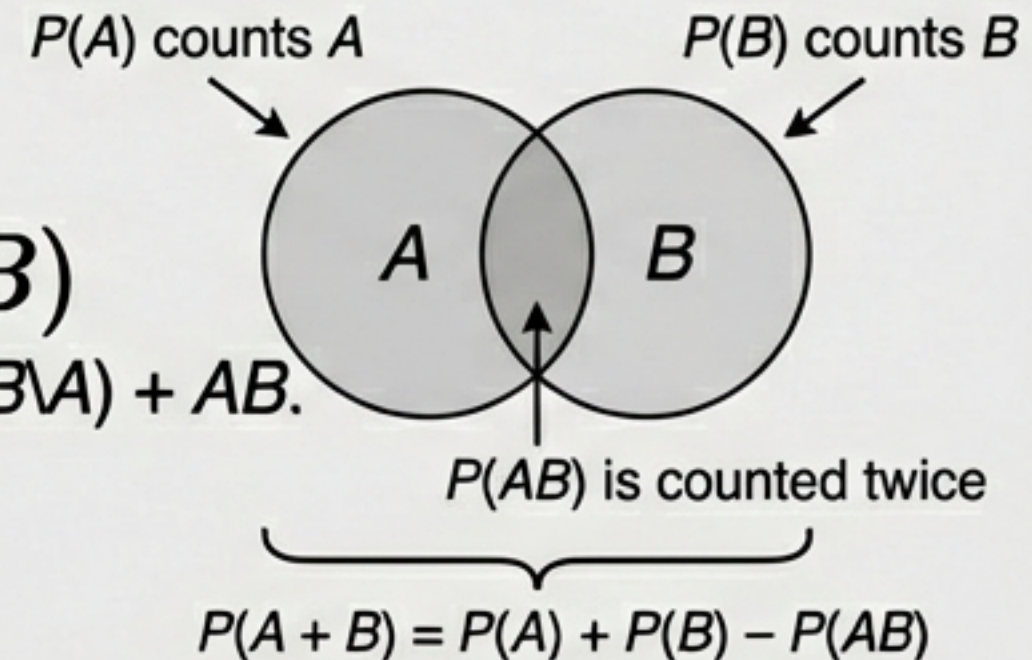
3. General Addition Rule

For any events, even if they overlap:

$$P(A + B) = P(A) + P(B) - P(AB)$$

We can write $A + B$ as a union of mutually exclusive events: $A + B = (A \setminus B) + (B \setminus A) + AB$.

We also know $A = (A \setminus B) + AB$ and $B = (B \setminus A) + AB$. By Axiom III, $P(A \setminus B) = P(A \setminus B) + P(AB)$, and, $P(B) = P(B \setminus A) + P(AB)$. Substituting, $P(A \setminus B) + P(B \setminus A) + P(AB) = P(A + B) + P(AB)$. Rearranging gives the result.



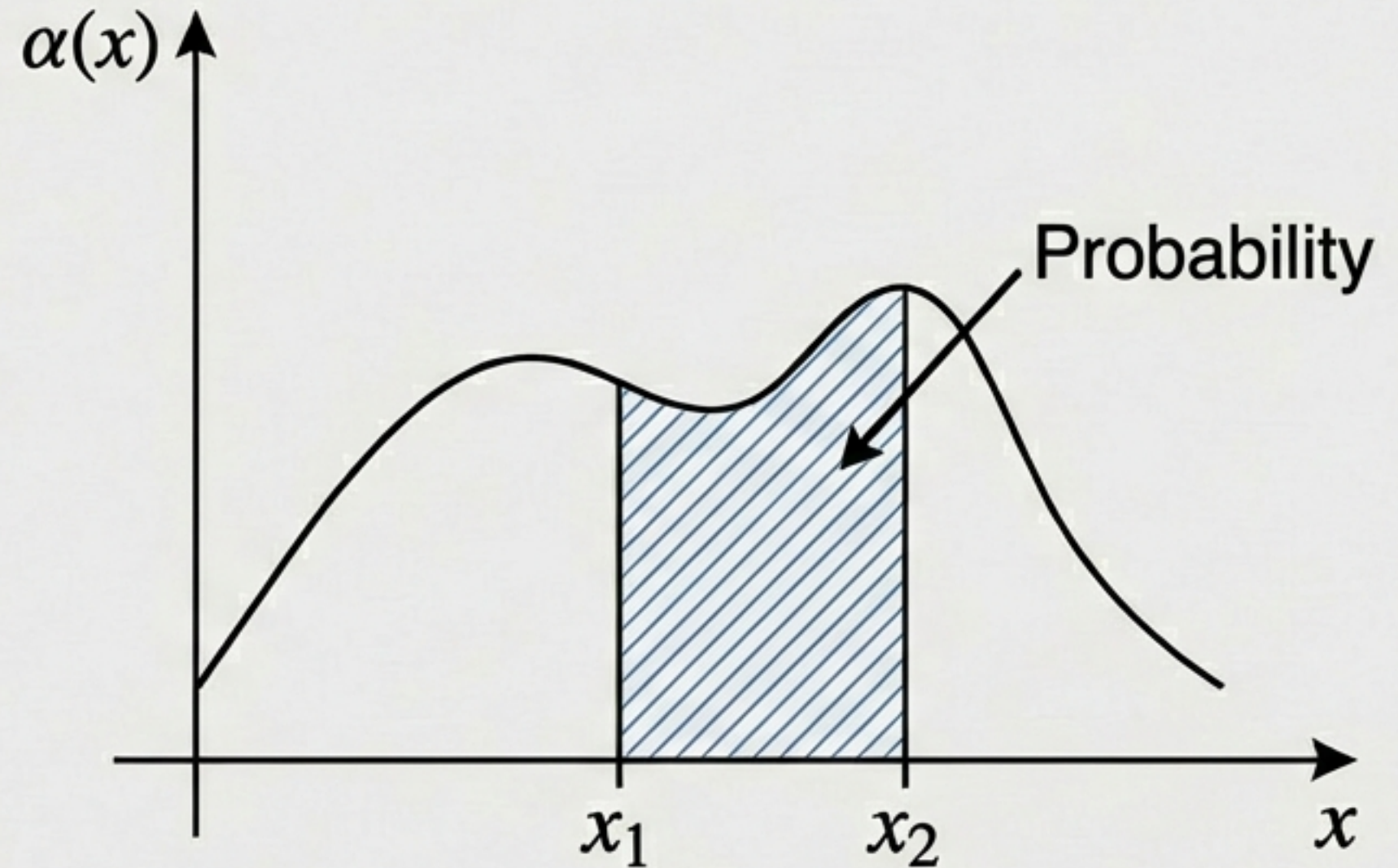
Probability on the Real Line

From counting outcomes to measuring area.

When the space S is continuous (like time or distance), the probability of a single point is zero.

We define probability using a Density Function $\alpha(x)$.

$$P\{x_1 < x \leq x_2\} = \int_{x_1}^{x_2} \alpha(x) dx$$

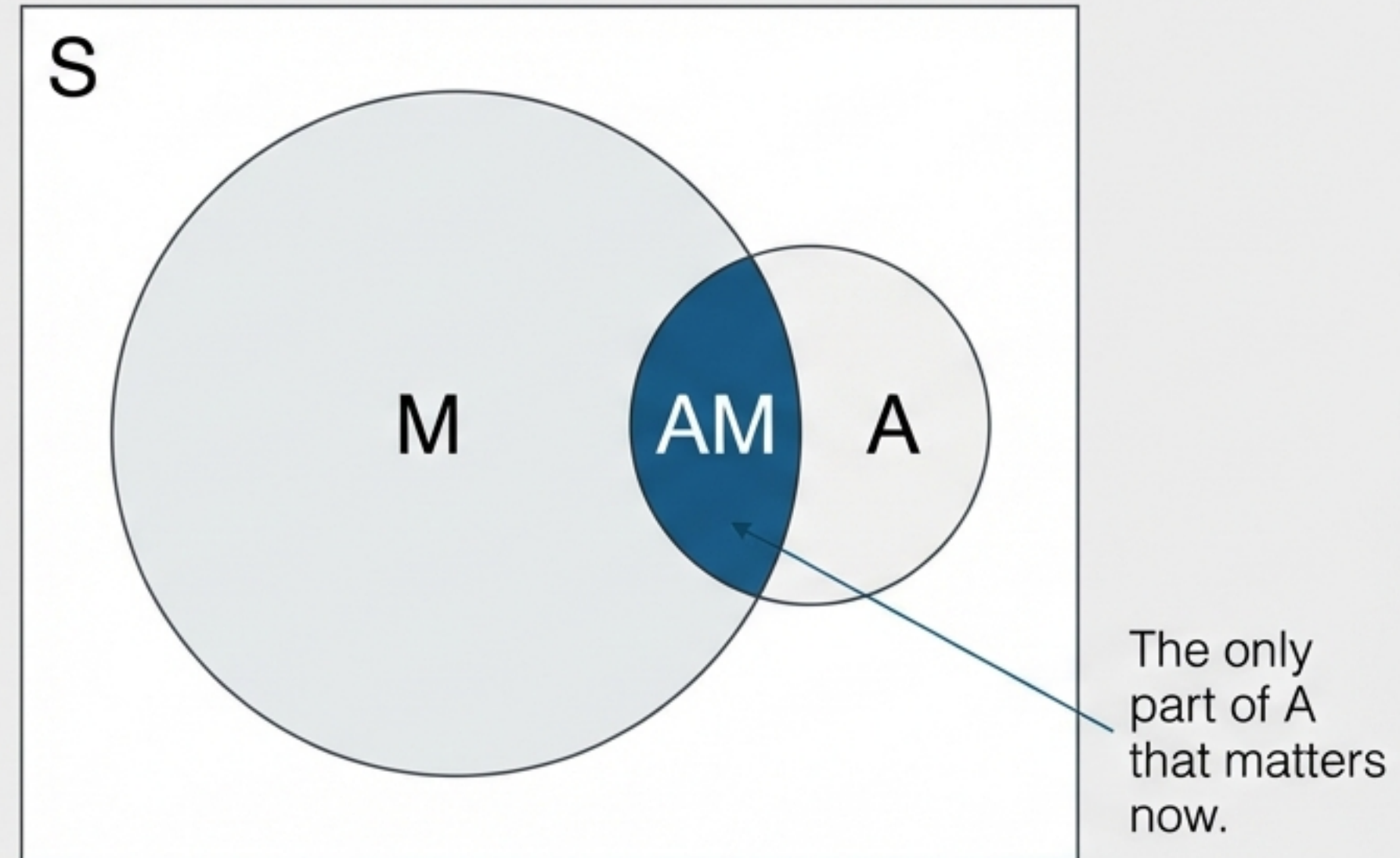


Conditional Probability

Renormalizing the Universe

When we assume event M has occurred, M becomes the new **Certain Event**.

$$P(A \mid M) = \frac{P(AM)}{P(M)}$$

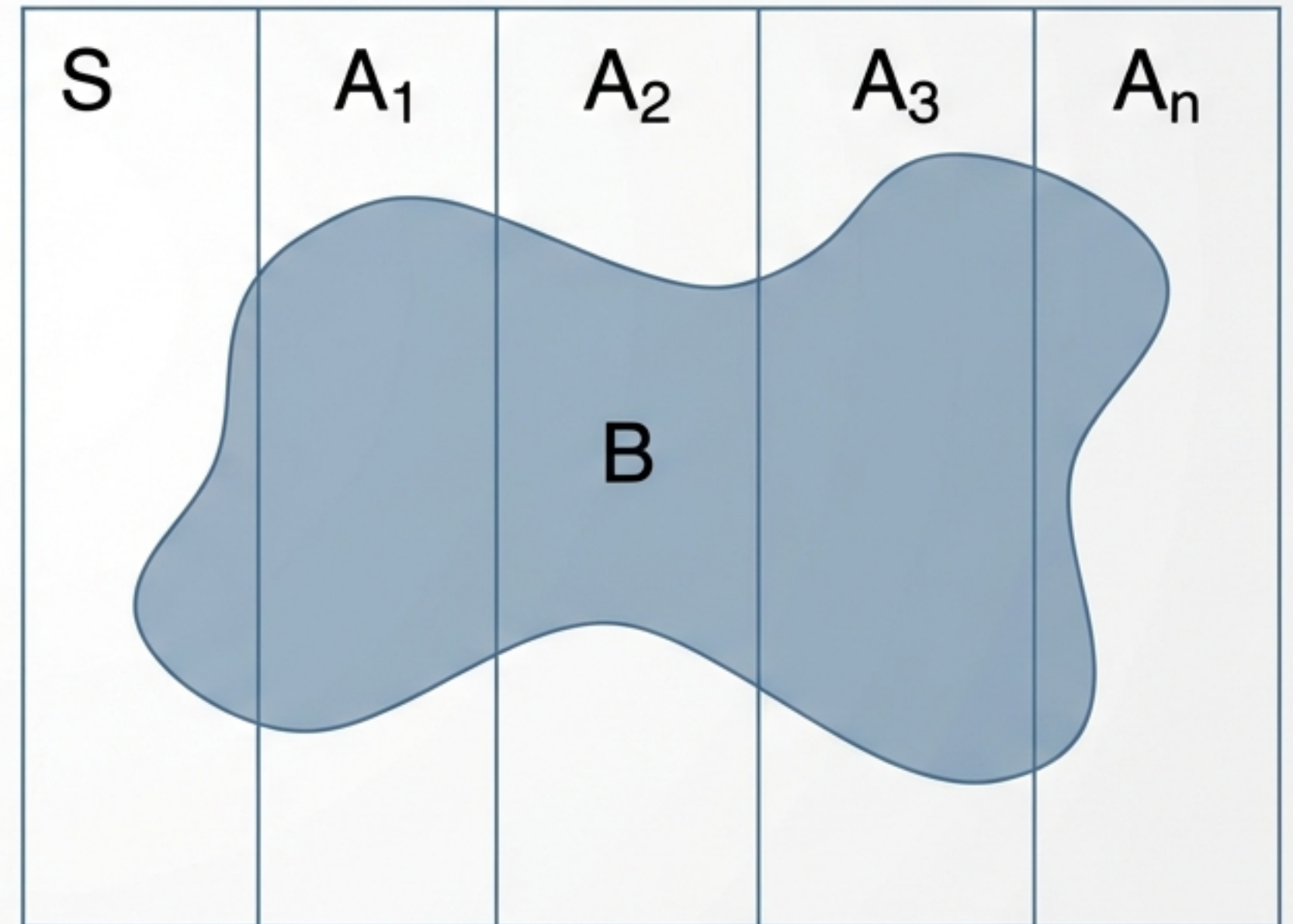


The Total Probability Theorem

Swiss Engineering Academic

To find $P(B)$, sum the weighted probabilities of the fragments:

$$P(B) = P(B|A_1)P(A_1) + \dots + P(B|A_n)P(A_n)$$



Bayes' Theorem: Reasoning Backward

We often know the effect (A) given the cause (B), but we need to infer the cause given the effect.

Step 1: Definition

$$P(B|A) = \frac{P(AB)}{P(A)}$$

Step 2: Substitution

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Step 3: The Theorem

$$P(\text{Cause} | \text{Effect}) = \frac{P(\text{Effect} | \text{Cause}) * P(\text{Prior Cause})}{P(\text{Total Probability of Effect})}$$

Example: The Defective Components



Experiment Flow:

1. Pick a box at random (Prior $P(\text{Box } 2) = 0.25$).
2. Pick a part. It is defective (Evidence).
3. Question: What is the probability it came from Box 2?

Calculation:

Using Bayes' Theorem:

$$P(\text{Box } 2 \mid \text{Defective}) = (0.4 * 0.25) / 0.1625 = 0.615$$

Conclusion:

Posterior Belief: 61.5%.

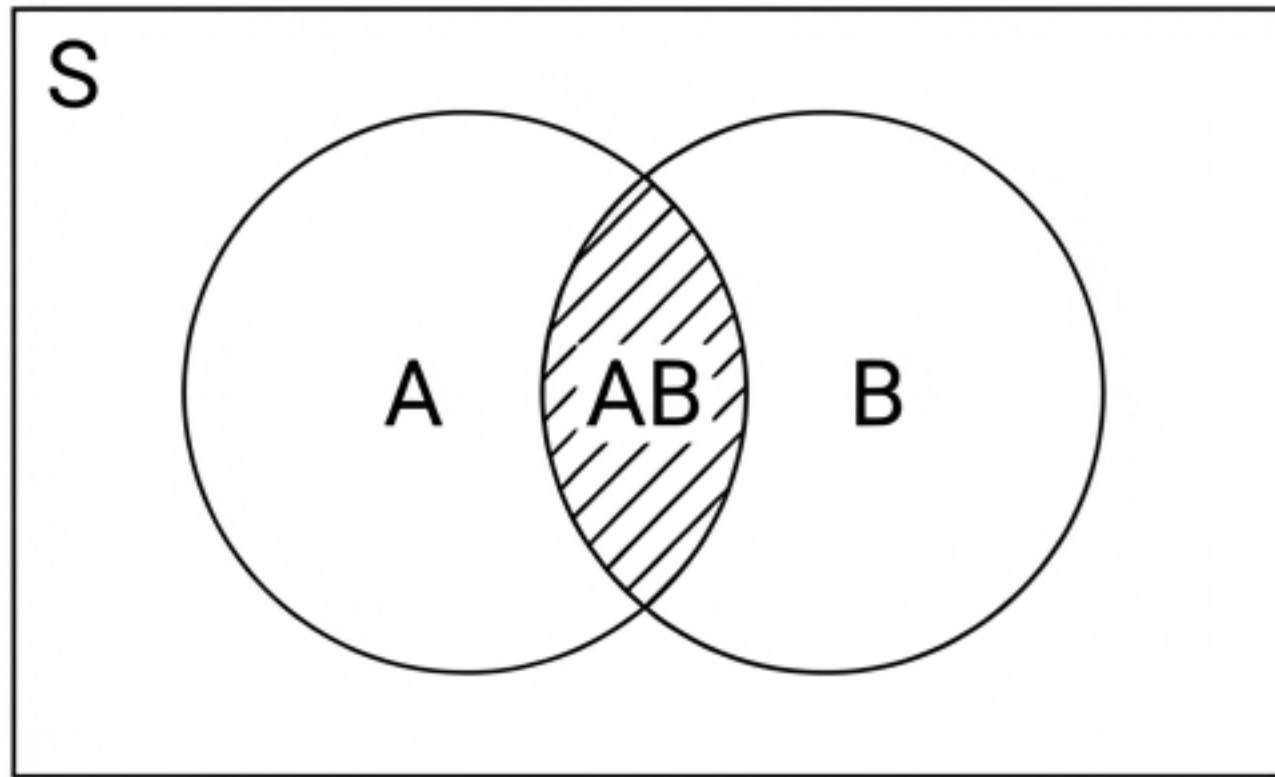
The evidence updated our belief from 25% to 61.5% because Box 2 had the highest defect rate.

Independence ≠ Mutually Exclusive

Independence

Definition: The occurrence of A provides NO information about B.

$$P(AB) = P(A)P(B)$$

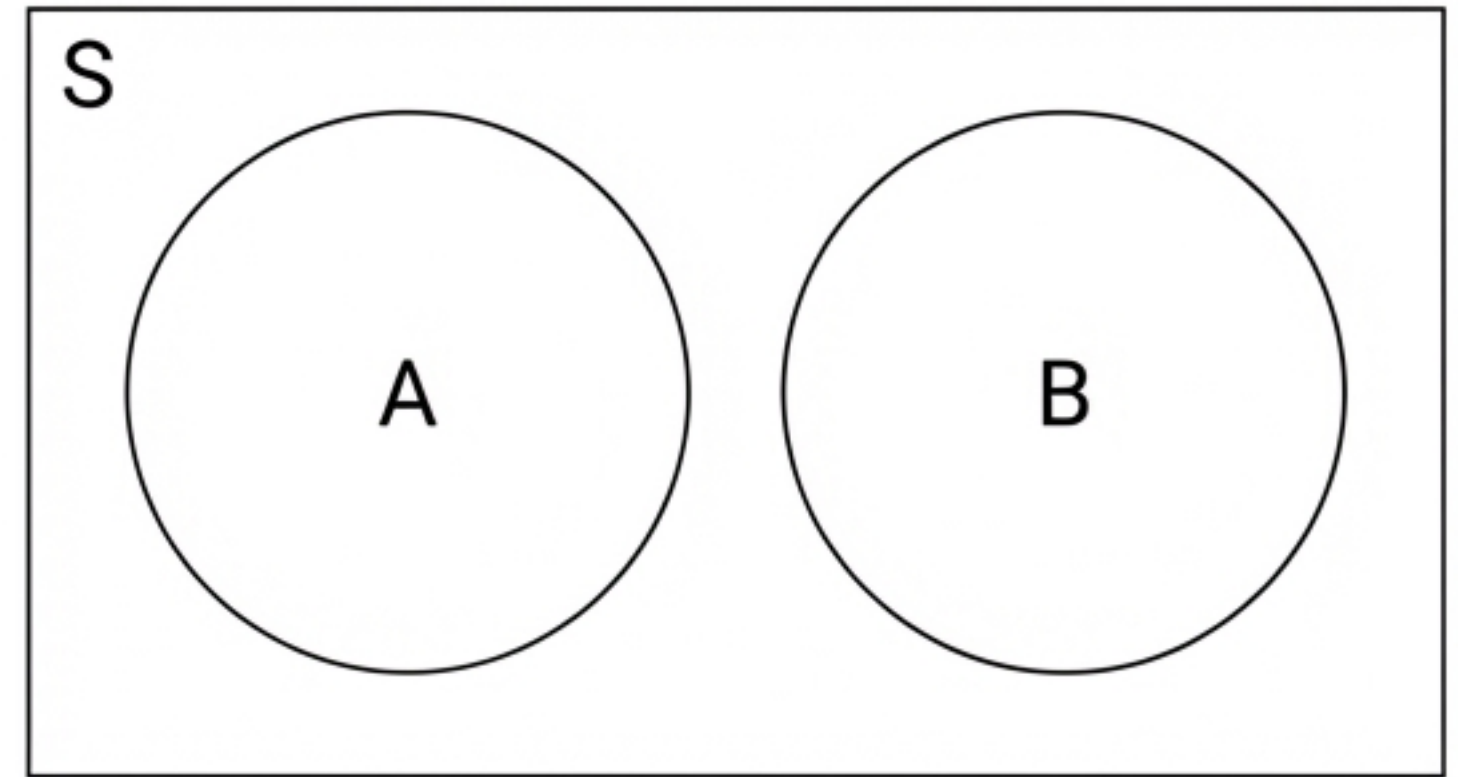


$$\text{Ratio(Overlap/A)} = \text{Ratio(B/S)}$$

Mutually Exclusive

Definition: If A occurs, B CANNOT occur.

$$AB = \emptyset$$



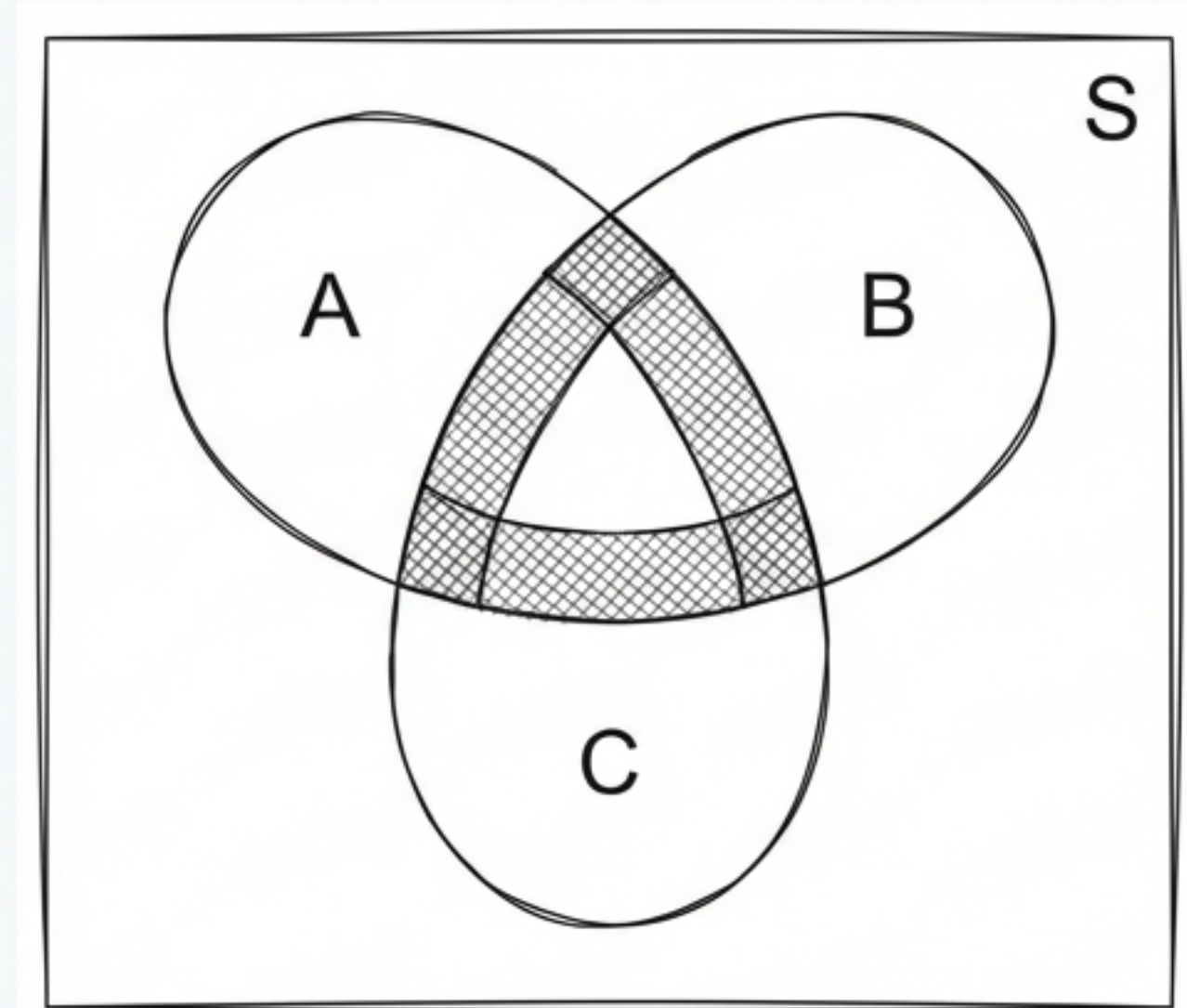
This is a form of extreme dependence. Knowing A tells you exactly what B is (it's not happening).

The Trap of Pairwise Independence

Independence in pairs does not guarantee mutual independence.

Condition for Mutual Independence requires 4 equations:

1. $P(AB) = P(A)P(B)$
2. $P(AC) = P(A)P(C)$
3. $P(BC) = P(B)P(C)$
4. $P(ABC) = P(A)P(B)P(C)$



Counter-Example: In the diagram shown, pairs are independent, but the trio is not.

The "Trefoil" Counter-Example

Combined Experiments: Cartesian Products

- Building complex experiments from simple ones.
- If S_1 has 6 outcomes (die) and S_2 has 2 outcomes (coin), the combined space S has 12 outcomes.

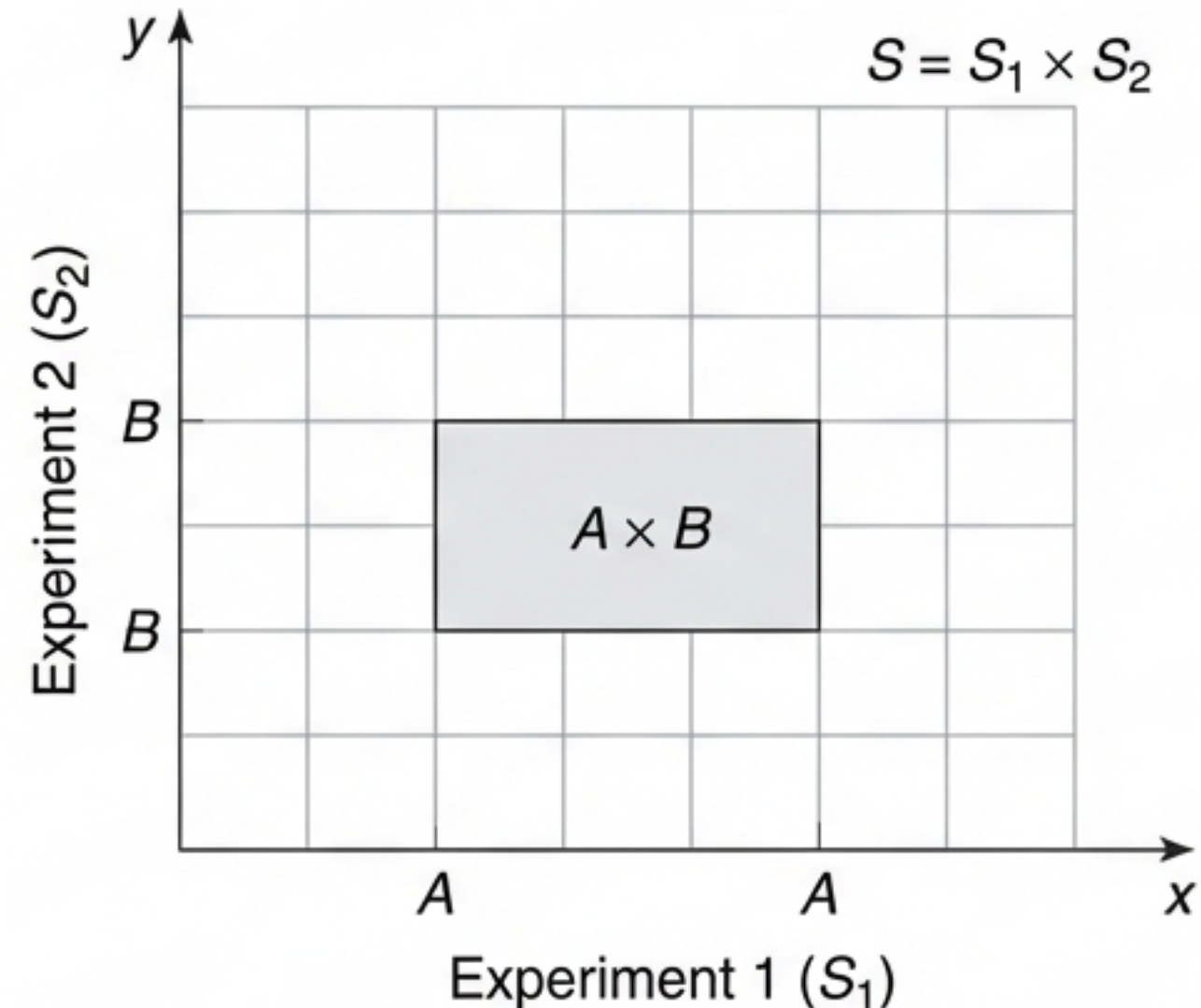


FIGURE 3-1 An event in the combined world is an intersection of an event from S_1 and an event from S_2 .

Repeated Trials and Sequences

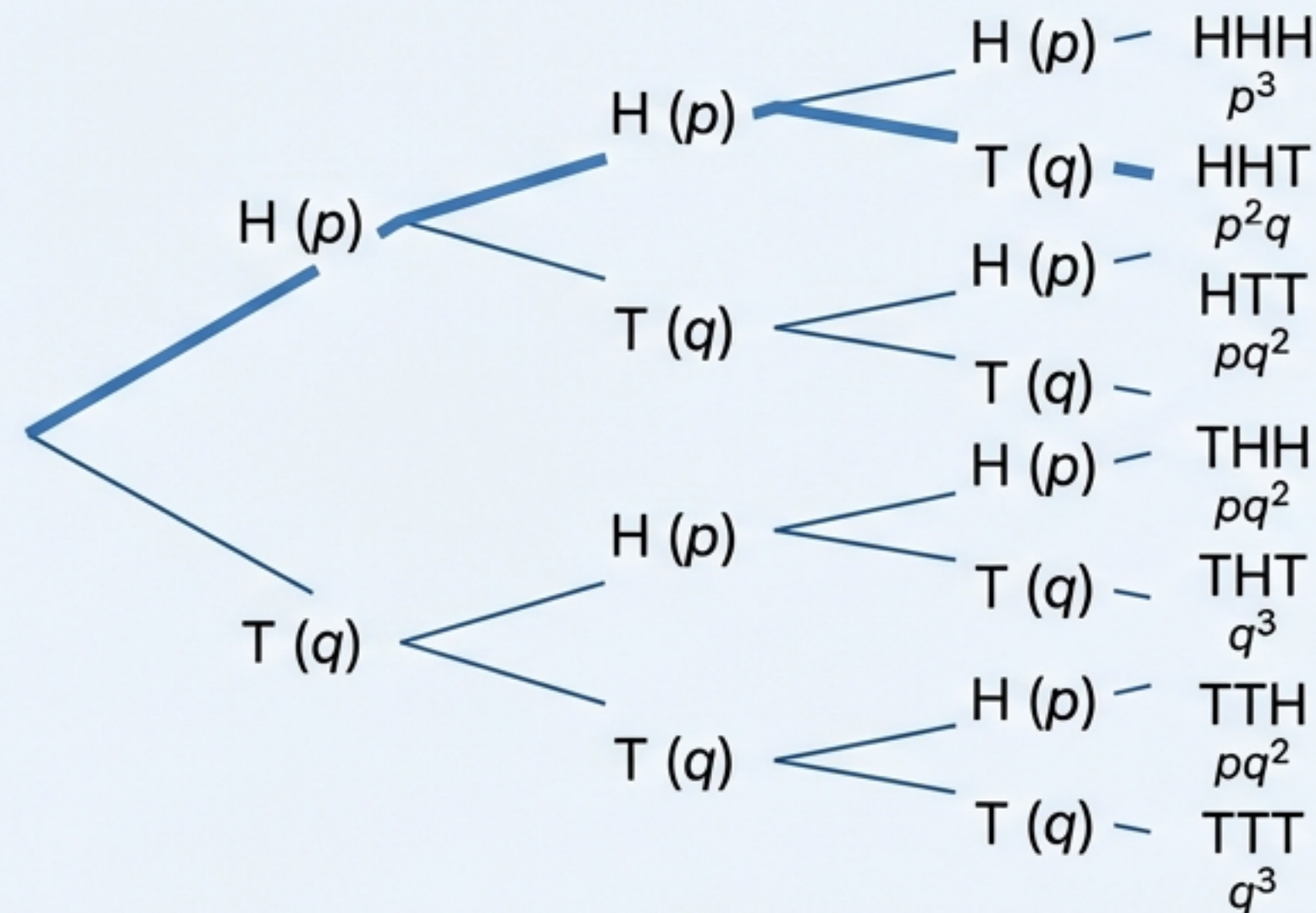
Independent Experiments Definition:

$$P(A \times B) = P(A)P(B)$$

Bernoulli Trials (Coin Tosses):

For a sequence of n trials with probability p :

$$P(k \text{ heads in } n \text{ tosses}) = \binom{n}{k} p^k \cdot (1 - p)^{n-k}$$



Causality versus Randomness

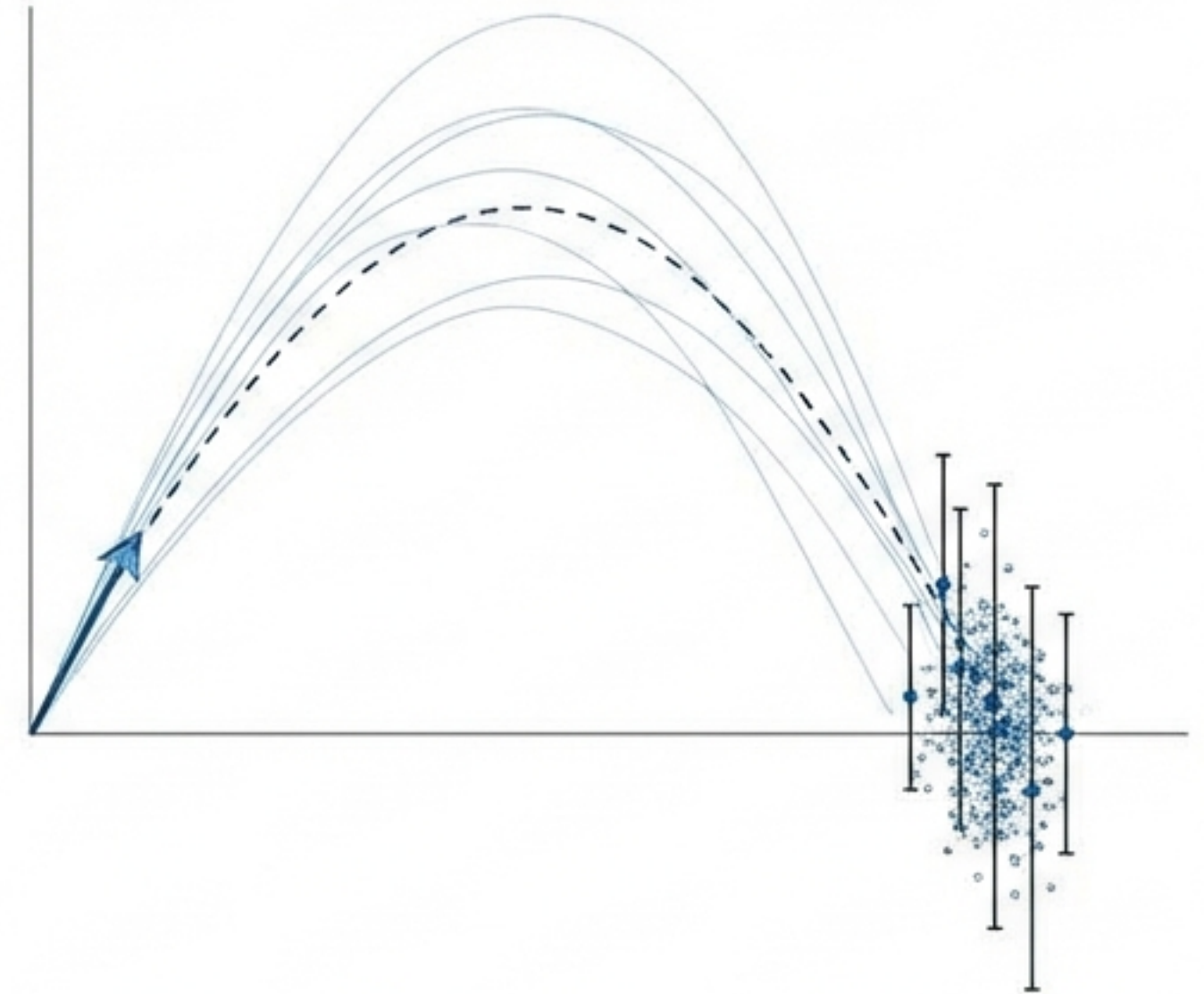
There is no conflict between determinism and probability.

Determinism

- Predicts the result of a single trial given exact parameters.
- Limited by our inability to measure the physical world perfectly.

Probability

- Predicts the averages of many trials.
- A rigorous logical structure to handle physical uncertainty.



"Scientific theories are not discoveries of the laws of nature but rather inventions of the human mind." - Papoulis